Software Engineering Lab

BCSE301P

Digital Assignment - 02

TOPIC: Local Services Finder

Submitted By:

VATSAL RAJ AGARWAL	23BCE0954
VEDANT KURLEKAR	23BCE0256
PRANAV KUMAR	23BCE0358
VAMSI KRISHNA	23BCI0024
CHERALATHAN M	23BCE0831

PROJECT SCHEDULING

Project scheduling for the Local Services Finder system involves systematically planning and organizing all development activities to ensure timely and efficient project completion. The project is divided into phases: Planning, Development, Enhancement, and Final Deployment.

The scheduling process begins with Requirement Analysis and UI/UX Wireframing, followed by core development tasks such as Authentication & Roles, Service Provider Module, and Customer Search & Booking. After implementing the primary functionalities, enhancement features like the Review & Rating System and Notifications & Alerts are developed.

Each task is assigned a specific duration and logical dependency to maintain workflow continuity. Milestones are defined to track progress, and overlapping activities are planned to optimize resource utilization. Finally, Testing & Debugging and Deployment & Optimization ensure quality assurance and successful system launch.

Overall, project scheduling ensures that the Local Services Finder is developed in an organized manner while maintaining timeline efficiency and balanced resource allocation.

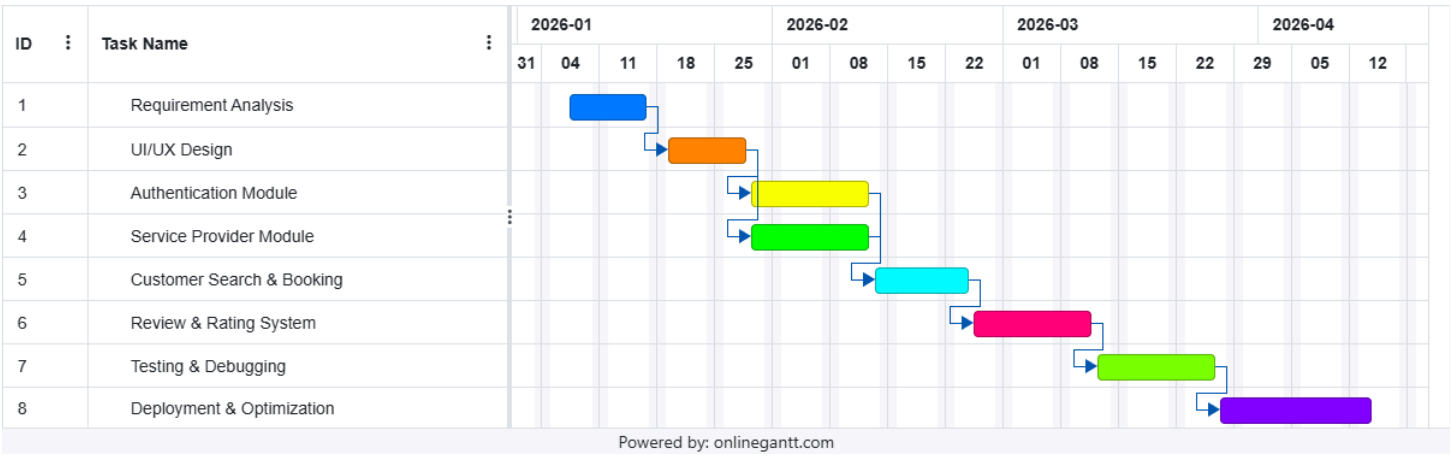
Gantt Chart

The Gantt chart represents the project timeline from January to April, showing the sequence and duration of tasks across different phases. The project starts with Requirement Analysis and UI/UX Wireframing during the Planning Phase.

In the Development Phase, core modules such as Authentication & Roles, Service Provider Module, and Customer Search & Booking are implemented. The Enhancement Phase includes features like Review & Rating System and Notifications & Alerts.

Finally, the project moves to Testing & Debugging, followed by Deployment & Optimization to ensure successful system launch. The chart clearly shows task overlap and major milestones, helping in tracking progress and timely completion of the Local Services Finder project.

ID	Task Name	Start	End	Duration	Dependency	Color
1	Requirement Analysis	2026-01-07	2026-01-16	8 days		
2	UI/UX Design	2026-01-19	2026-01-28	8 days	1FS	
3	Authentication Module	2026-01-29	2026-02-12	11 days	2FS	
4	Service Provider Module	2026-01-29	2026-02-12	11 days	2FS	
5	Customer Search & Booking	2026-02-13	2026-02-24	8 days	3FS,4FS	
6	Review & Rating System	2026-02-25	2026-03-11	11 days	5FS	
7	Testing & Debugging	2026-03-12	2026-03-26	11 days	6FS	
8	Deployment & Optimization	2026-03-27	2026-04-14	13 days	7FS	



TOOL USED: ONLINE GANTT

https://www.onlinegantt.com/#/gantt

Pert chart

The PERT (Program Evaluation and Review Technique) chart for the *Local Services Finder* project visually represents the sequence of activities, their dependencies, and time estimates required to complete the system. It helps in identifying the critical path and estimating the overall project duration.

In our project, the PERT network begins with Project Start, followed by Requirement Analysis (10 days) and UI/UX Design (11 days). After design completion, development branches into two parallel modules: Authentication Module (14 days) and Service Provider Module (14 days).

The Authentication Module proceeds to Customer Search & Booking (10 days), while the Service Provider Module connects to the same stage with zero additional delay. This is followed by Review & Rating System (14 days) and finally Deployment & Optimization (9 days).

The critical path highlighted in the diagram is:

Project Start → Requirement Analysis → UI/UX Design → Authentication Module → Customer Search & Booking → Review & Rating System → Deployment & Optimization

The total estimated project duration is 68 days.

Time Estimation Formula Used in PERT

PERT considers three time estimates for each activity:

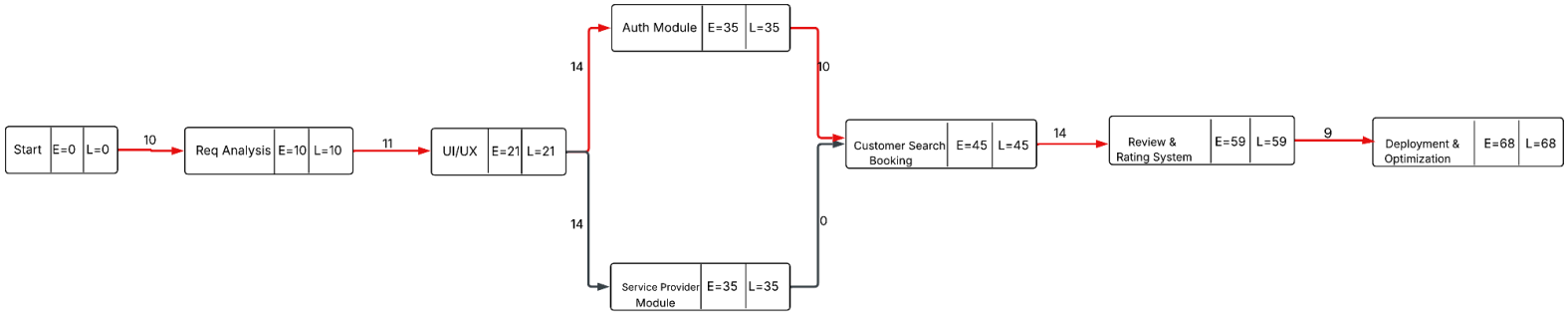
- **To** = Optimistic Time
- **Tm** = Most Likely Time
- **Tp** = Pessimistic Time

The expected time (T) is calculated using:

$$T = (Tp + 4Tm + To) / 6$$

This method helps in realistic duration estimation and risk analysis by considering best-case, most-likely, and worst-case scenarios.

Task	Activity Name	Optimistic (TO)	Most Likely (TM)	Pessimistic (TP)	Expected Time (TE)
A	Requirement Analysis	8	10	14	10
B	UI/UX Design	9	11	16	11
C	Authentication Module	12	14	18	14
D	Service Provider Module	12	14	20	14
E	Customer Search & Booking	12	14	20	14
F	Review & Rating System	8	10	14	10
G	Testing & Debugging	12	14	18	14
H	Deployment & Optimization	6	9	12	9



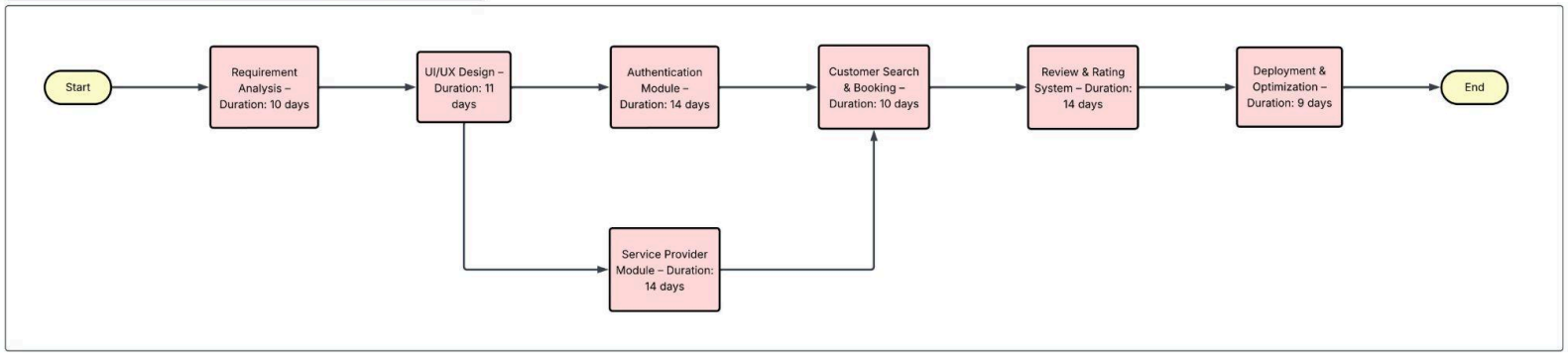
Tool Used: Lucid Chart [Lucid Chart Link]

Activity Network Model

Activity Network Model for the Local Services Finder project visually represents the sequence and dependency of tasks involved in developing the system. The project begins with Requirement Analysis (10 days), followed by UI/UX Design (11 days). After the design phase, development branches into two parallel activities: Authentication Module (14 days) and Service Provider Module (14 days). Both modules must be completed before proceeding to Customer Search & Booking (10 days). This is followed by the Review & Rating System (14 days) and finally Deployment & Optimization (9 days), leading to project completion.

The Activity Network Model helps in identifying the critical path, which is Requirement Analysis → UI/UX Design → Authentication Module → Customer Search & Booking → Review & Rating System → Deployment & Optimization, with a total project duration of 68 days. By visualizing task dependencies and parallel activities, the model supports efficient scheduling, better resource allocation, and timely achievement of project milestones. It ensures that key development activities are properly coordinated, contributing to the organized and successful completion of the Local Services Finder prototype.

Activity Network Diagram – Local Services Finder



Tool Used: Lucid Charts [Lucid Chart Link]

Critical Path Method

The Critical Path Method (CPM) is a vital project management technique used in the development of the Local Services Finder prototype. In our project, CPM helps identify the sequence of activities that determine the minimum time required to complete the system. Tasks such as Requirement Analysis, UI/UX Design, Authentication Module development, Customer Search & Booking, Review & Rating System, and Deployment & Optimization are interconnected and form the critical path of the project.

By calculating earliest and latest start times, CPM allows us to determine which activities have zero slack and directly impact the overall project duration of 68 days. When combined with the Gantt chart and PERT chart, CPM provides a complete overview of the project schedule. The Gantt chart visually represents task timelines and overlaps, while the PERT chart helps estimate realistic activity durations. Together, these tools improve timeline management, optimize resource utilization, and ensure the timely and successful delivery of the Local Services Finder prototype.

RED arrow represents the CRITICAL PATH

Tool Used: Lucid Chart Link

Staff Allocation

In the Local Services Finder project, responsibilities are distributed among team members based on their technical strengths, role suitability, and development phase involvement. This structured allocation ensures balanced workload, accountability, and efficient execution of frontend modules, UI design, booking logic, and deployment activities.

Task	Task Name	Pranav Kumar	Vatsal Raj Agarwal	Cheralathan M	Kurlekar Vedant Anand	Vamsi Krishna P
A	Project Planning	High	High		Moderate	Moderate
B	Requirement Analysis	High		Moderate	High	
C	UI/UX Design (Figma)	Moderate	Moderate			High
D	Authentication & Roles	High			High	
E	Service Provider Module		High	Moderate		Moderate
F	Customer Search & Booking	High	High		Moderate	
G	Review & Rating System		Moderate	High		Moderate
H	Testing & Deployment	Moderate	High	Moderate	High	

Cost Estimation

COCOMO MODEL

Cost estimation for the Local Services Finder project was performed using the COCOMO (Constructive Cost Model). This model estimates the development effort, project duration, and team size based on the size of the software measured in KLOC (Thousands of Lines of Code) and various cost drivers.

To improve estimation accuracy, an Effort Adjustment Factor (EAF) was calculated using selected cost drivers such as developer capability, experience, use of tools, and software engineering practices.

S/W project	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Estimated KLOC:

Total (Frontend + Backend) = 7 KLOC (approx)

Cost drivers:

Storage constraints: Low = 1

Required reliability: Nominal = 1

Database size: Nominal = 1

Product complexity: Nominal = 1

Experience in developing similar software: High = 0.91

Programming capabilities of the developers: High = 0.86

Analyst capability: High = 0.91

Application of software engineering methods: High = 0.91

Use of software tools: High = 0.91

Required development schedule: Nominal = 1

All other cost drivers: Nominal = 1

Basic COCOMO Formula

Effort = a(KLOC)^b
Time = c(Effort)^d
Persons Required = Effort / Time
EAF=0.91×0.86×0.91×0.91×0.91 = 0.59(approx)

Project Type: Semi-Detached

Suitable for medium-complexity projects.

Effort: a(KLOC)^b
3(7)^1.12
26.52 PM

Time : c(Effort)^d
2.5(26.52)^0.35
7.87 Months

Persons Required : Effort/Time
26.52/7.87
3.37 (4 approx) persons

Effort Corrected = Effort * EAF =26.52 * 0.59 = 15.65PM

Time : c(Effort Corrected)^d = 2.5(15.65)^0.35 = 6.55 Months

Project Type: Organic

Suitable for small, simple projects with experienced teams.

Effort: a(KLOC)^b
2.4(7)^1.05
18.5 PM

Time: c(Effort)^d
2.5(18.5)^0.38
7.57 months

Persons Required : Effort/Time
18.5/7.57
2.44 (3 approx) persons
Effort Corrected = Effort * EAF =18.5 * 0.59 = 10.92PM

Time : c(Effort Corrected)^d = 2.5(10.92)^0.38 = 6.2 Months

Project Type: Embedded

Suitable for highly complex systems with strict constraints.

Effort: a(KLOC)^b
3.6(7)^1.20
37.19 PM

Time : c(Effort)^d
2.5(37.19)^0.32
7.95 Months

Persons Required : Effort/Time
37.19/7.95
4.68 (5 approx) persons

Effort Corrected = Effort * EAF =37.19 * 0.59 = 21.95PM

Time : c(Effort Corrected)^d = 2.5(21.95)^0.32 = 6.72 Months